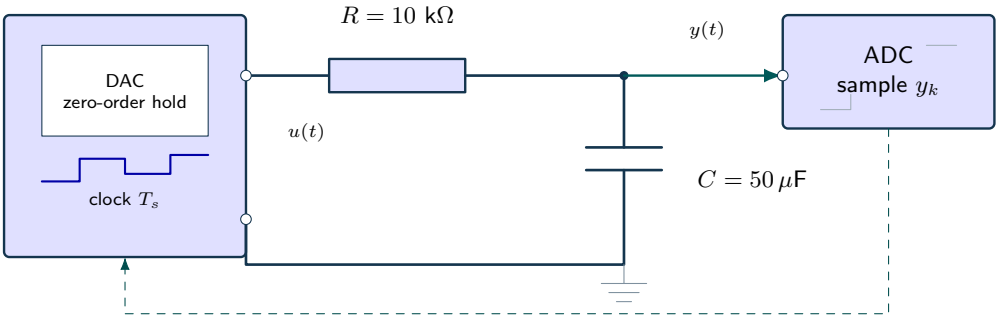

Continuous energy store

$RC = 0.5$ s; $\dot{y} = 2(u - y)$.
The held voltage drives a real-valued continuous state between samples.

Clocked interface

$y_k = y(kT_s)$.
$u(t) = u_k$ for $kT_s \le t < (k+1)T_s$.
Clock and update order matter.

Ideal converter boundary

No quantization, electrical loading or converter dynamics. Use exact held-input discretization when comparing sample periods.